

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Revision Test 2026 — ANSWER KEY (Version 1)

Chemistry — Class 9 (Science Group)

Syllabus: Chapter 1 - Nature of Chemistry in Science | Chapter 2 - Matter | Chapter 3 - Atomic Structure | Chapter 4 - Periodic Table & Periodicity of Properties | Chapter 5 - Chemical Bonding | Chapter 6 - Stoichiometry | Chapter 7 - Electrochemistry | Chapter 8 - Energetics | Chapter 9 - Chemical Equilibrium

Total Marks: 71

Time: 2 Hours 15 Minutes

Paper: Exam-20 (Version 1)

SECTION A — MCQ Answer Key (60 × 1 = 60 Marks)

#	Question	Correct answer	Explanation	Source in her textbook
1	Which branch of chemistry investigates the essential question: "How do biomolecules such as carbohydrates, proteins, nucleic acids and lipids contribute to the structure and function of living organisms?"	C. biochemistry 1	Section 1.3 lists this exact question under Biochemistry, the branch that studies the physical and chemical changes taking place in living organisms.	Ch1 §1.3; SLO 3 (formulate examples of essential questions important for the branch of chemistry)
2	"How do geological processes influence the distribution of elements in the Earth's crust?" is an essential question of:	C. geochemistry 1	Geochemistry deals with the chemical composition, distribution and transformation of elements and compounds in the Earth's crust — its rocks, minerals, soils, water and atmosphere.	Ch1 §1.3; SLO 3 (essential questions of a branch)
3	"How do nuclear reactions differ from chemical reactions, and what are their applications?" is an essential question of:	D. nuclear chemistry 1	Nuclear chemistry studies the changes that occur in atomic nuclei. Section 1.3 also lists "What is the role of radioisotopes in medicine and industry?" under this branch.	Ch1 §1.3; SLO 3 (essential questions of a branch)
4	The branch of chemistry that deals with the chemical processes and reactions occurring in astronomical environments such as stars, planets and comets is:	B. astrochemistry 1	Astrochemistry is defined in Section 1.2 as the branch dealing with chemical processes and reactions that occur in astronomical environments, including stars, planets, comets and interstellar space.	Ch1 §1.2; SLO 2 (chemistry has many sub-fields and interdisciplinary fields)
5	A solution of any metal in liquid mercury is called:	B. an amalgam 1	Under solid solutions, Section 2.4 gives "Amalgam — solution of any metal in liquid mercury". An alloy is a solution of one solid metal in another, such as brass or bronze.	Ch2 §2.4; SLO (identify solutions, colloids and suspensions as mixtures and give an example of each)
6	Fog, clouds and mist are solutions in which the solute is liquid water and the solvent is:	A. a gas 1	Section 2.4 lists fog, clouds and mist as gaseous solutions in which liquid water is the solute and air, a gas, is the solvent.	Ch2 §2.4; SLO (identify solutions as mixtures and give an example)
7	In the ghee industry hydrogen gas is dissolved on the surface of solid nickel. This mixture is an example of:	A. a solid solution 1	The state of the SOLVENT decides the type of solution. Here nickel, the solvent, is a solid, so the mixture is a solid solution even though the solute is a gas.	Ch2 §2.4; SLO (identify solutions as mixtures and give an example)
8	The solubility of a solute depends on the solvent used, the temperature and the:	C. pressure 1	Section 2.4.7 states that solubility depends on the solvent used, temperature and pressure.	Ch2 §2.4.7; SLO (explain the effect of temperature on solubility and formation of unsaturated and saturated solutions)

9	A carbonated drink, in which carbon dioxide is dissolved in water, is an example of:	B. a gas dissolved in a liquid 1	Section 2.4 lists carbonated drinks under liquid solutions: carbon dioxide gas is the solute and liquid water is the solvent.	Ch2 §2.4; SLO (identify solutions as mixtures and give an example)
10	An atomic model is best described as:	B. a simplified representation that helps us to understand the structure of an atom 1	Section 3.1 states that an atomic model is a conceptual tool, a simplified representation of a complex reality, used as an aid to understand the structure of an atom.	Ch3 §3.1; SLO (explain that an atomic model is an aid to understand the structure of an atom)
11	Tritium is not found in naturally occurring hydrogen because its nucleus is:	A. highly unstable 1	Section 3.5.1 states that tritium is radioactive and rare, and is not found in naturally occurring hydrogen because its nucleus is highly unstable.	Ch3 §3.5.1; SLO (state that isotopes can exhibit radioactivity)
12	Radioisotopes are isotopes that have:	D. an unstable combination of neutrons and protons 1	Section 3.2 defines radioisotopes as isotopes having an unstable combination of neutrons and protons, or excess energy in their nucleus. Examples given are hydrogen-3, carbon-14 and uranium-238.	Ch3 §3.2; SLO (state that isotopes can exhibit radioactivity)
13	The number of neutrons in one atom of the isotope ^{80}Br (atomic number 35) is:	D. 45 1	Number of neutrons = mass number – atomic number = $80 - 35 = 45$.	Ch3 §3.3; SLO (determine the number of protons and neutrons of different isotopes)
14	When a nucleus emits a proton during radioactive decay, the atom is changed into:	B. another element 1	Section 3.3 explains that emitting a neutron changes the atom into another isotope, but emitting a proton changes the proton number, so it becomes another element altogether.	Ch3 §3.3; SLO (state that radioactivity can change the proton number and alter an atom's identity)
15	The strong nuclear force that binds the nucleus together acts between:	D. proton and proton, neutron and neutron, and neutron and proton 1	Section 3.1 states that the nuclear force exists between neutron and neutron, proton and proton, and neutron and proton. It is much stronger than electrostatic or magnetic forces.	Ch3 §3.1; SLO (explain that nucleus is made up of protons and neutrons held together by strong nuclear force)
16	Protons and electrons moving at the same speed are passed through a uniform electric field. The electrons are deflected far more strongly than the protons because the electron's:	A. mass is very much smaller 1	Section 3.2 notes that at the same speed the much lighter electrons are deflected far more strongly than the heavier protons. Protons bend towards the negative plate and electrons towards the positive plate.	Ch3 §3.2; SLO (illustrate the path that positively and negatively charged particles would take under the influence of a uniform electric field)
17	In Rutherford's planetary model of the atom, the electrostatic force of attraction between the nucleus and a revolving electron is balanced by the:	A. centrifugal force of the revolving electron 1	Section 3.1 states that in Rutherford's model the centrifugal force of the revolving electrons balances the electrostatic force of attraction, which keeps the electron in its orbit.	Ch3 §3.1; SLO (explain that an atomic model is an aid to understand the structure of an atom)
18	According to the Auf Bau principle, the sub-shell filled immediately after the 3p sub-shell is:	B. 4s 1	The increasing order of energy is $1s < 2s < 2p < 3s < 3p < 4s < 3d$. Electrons fill the lowest available energy sub-shell first, so 4s fills before 3d.	Ch3 §3.7; SLO (interpret and use electronic configuration; Auf Bau principle)
19	An atom forms an anion by gaining one or more electrons. This happens most readily when the atom has a relatively high:	D. electron affinity 1	Section 3.6 states that the formation of an anion occurs when an atom has a relatively high electron affinity, that is, a	Ch3 §3.6; SLO (describe the formation of positive (cation) and negative

			strong tendency to accept electrons.	(anion) ions from atoms)
20	The block to which an element belongs is decided by:	B. the sub-shell in which its last electron is placed 1	Section 4.1.1 states that the block of an element is the orbital (sub-shell) in which the last electron is placed: s, p, d or f.	Ch4 §4.1.1; SLO (identify the group or period or block of an element using its electronic configuration)
21	An element whose electronic configuration is $1s^2 2s^2 2p^2$ belongs to:	C. Group 14 1	The last electron enters a p sub-shell, so the element is in the p block. Valence electrons = $2 + 2 = 4$, and for the p block the group number = valence electrons + 10 = 14.	Ch4 §4.1.1; SLO (identify the group or period or block of an element using its electronic configuration)
22	The elements of Group 16 of the periodic table are collectively known as the:	C. chalcogens 1	Section 4.2 names the Group 16 elements (oxygen, sulphur, selenium) the chalcogens. They have six valence electrons and gain two electrons to form divalent anions.	Ch4 §4.2; SLO (use terms like alkali metals, alkaline earth metals, halogens, noble gases ... in reference to the periodic table)
23	Across a period of the periodic table, the density of the elements:	A. increases to a maximum near the middle of the period and then decreases 1	Section 4.4.3 states that across a period density tends to increase from left to right up to a maximum around the middle of the period, and then decreases.	Ch4 §4.4.3; SLO (identify trends in groups and periods ... including density)
24	The first ionization energies of four elements are Na = 496, Mg = 738, Al = 578 and Si = 786 kJ/mol. The element that holds its outermost electron most tightly is:	D. Si 1	A high ionization energy means a stronger attraction between the nucleus and the outermost electron. Silicon has the largest value, 786 kJ/mol, so it holds its outermost electron most tightly.	Ch4 §4.3.3, Fig. 4.5; SLO (identify trends ... including ionization energy)
25	The hydrogen halide with the greatest thermal stability is:	D. HF 1	The larger the electronegativity difference between hydrogen and the halogen, the stronger the H-X bond. That difference is greatest for fluorine, so HF has the strongest bond and the greatest thermal stability. Stability falls from HF to HI.	Ch4 §4.7; SLO (analyze the relative thermal stabilities of the hydrogen halides and explain these in terms of bond strengths)
26	Transition elements have high melting points mainly because of:	C. stronger metallic bonding arising from partially filled d sub-shells 1	Section 4.5 states that transition elements have high melting points due to stronger metallic bonding, which arises because the partially filled d sub-shells also take part in bonding.	Ch4 §4.5; SLO (describe the transition elements as metals that have high densities, high melting points ...)
27	The noble gases are present in the atmosphere as:	D. mono-atomic gases 1	Section 4.8 defines the Group 18 noble gases as un-reactive, odourless, colourless mono-atomic gases whose very low reactivity is due to a complete valence shell.	Ch4 §4.8; SLO (define the Group 18 noble gases as un-reactive mono-atomic gases)
28	All of the halogens exist as:	C. diatomic molecules 1	Section 4.7 states that all the halogens are reactive non-metals which exist as diatomic molecules: F_2 , Cl_2 , Br_2 and I_2 .	Ch4 §4.7; SLO (define Group VII halogens as diatomic non-metals)
29	Reactivity increases on going down Group 1, so the least reactive alkali metal is:	B. lithium 1	Section 4.4 states that lithium, at the top of the group, is the least reactive alkali metal, and that sodium is more reactive than lithium. Reactivity increases down the group.	Ch4 §4.4; SLO (predict properties of elements in Group 1 in order of reactivity given relevant information)
30	The period number of an element is the	C. the principal	Section 4.1.1 states that the	Ch4 §4.1.1; SLO

	same as:	quantum number (n) of its valence shell 1	period of an element is the principal quantum number (n) of the valence electron. For example sodium, $1s^2 2s^2 2p^6 3s^1$, has $n = 3$ and lies in period 3.	(identify the group or period or block of an element using its electronic configuration)
31	Elements whose atoms have five, six or seven valence electrons usually react by:	B. gaining electrons to form anions 1	Section 4.2 states that elements with 5, 6 or 7 valence electrons tend to gain 3, 2 or 1 electrons respectively, forming ions of charge -3 , -2 and -1 .	Ch4 §4.2; SLO (explain the relationship between group number and the charge of ions formed ...)
32	Non-metals generally have lower melting and boiling points than metals because their atoms and molecules are held together by:	A. weaker forces such as covalent, van der Waals or hydrogen bonds 1	Section 4.9 states that metals have high melting and boiling points due to strong metallic bonds, while non-metals have lower values because their atoms and molecules are held by weaker bonds such as covalent, van der Waals or hydrogen bonds.	Ch4 §4.9; SLO (compare the general physical properties of metals and non-metals ... melting points and boiling points)
33	Carbon dioxide contains strong covalent bonds inside each molecule, yet it has a very low melting and boiling point. This is because:	C. the forces between its separate molecules are weak 1	Section 5.5 contrasts CO_2 with silicon dioxide. CO_2 has strong covalent bonds between C and O atoms but only weak intermolecular forces between molecules, so little energy is needed to separate the molecules.	Ch5 §5.5; SLO (interpret the strength of forces of attraction and their impact on melting and boiling points)
34	In a molecule of hydrogen cyanide, HCN, the bond joining the carbon atom to the nitrogen atom is a:	B. triple covalent bond 1	Section 5.3.2 explains that in HCN carbon shares one electron with hydrogen to form a single bond and shares three electrons with nitrogen, forming a triple covalent bond.	Ch5 §5.3.2; SLO (define ionic, covalent, coordinate covalent and metallic bonds)
35	Boron trifluoride, BF_3 , is described as electron deficient because the boron atom has only:	B. three pairs of electrons in its valence shell 1	Section 5.3.4 states that boron in BF_3 has only three pairs of electrons in its valence shell and needs one more pair to complete its octet, which is why it is called electron deficient.	Ch5 §5.3.4; SLO (define ionic, covalent, coordinate covalent and metallic bonds)
36	According to the textbook, the adhesive action of paints and dyes is developed because of:	C. hydrogen bonding 1	The Key Points of Chapter 5 state that the adhesive action of paints and dyes is developed due to hydrogen bonding. Epoxy adhesives contain partially positive hydrogen atoms and oxygen atoms carrying lone pairs.	Ch5 §5.4, Key Points; SLO (explain the properties of compounds in terms of bonding and structure)
37	The two-dimensional layers of graphite are held to one another by:	A. weak van der Waals forces 1	Section 5.5.1 states that each carbon atom in a layer is joined to three others by single covalent bonds, and the fourth valency is satisfied by weak van der Waals forces between the layers, which is why the layers slide over each other.	Ch5 §5.5.1; SLO (justify the suitability of usage of graphite, diamond and metals for industrial purposes)
38	A beryllium atom, $1s^2 2s^2$, loses two electrons. The ion formed has the electronic configuration of:	C. helium 1	Be loses its two 2s electrons to form Be^{2+} , which is left with $1s^2$. That is the configuration of helium, so beryllium obeys the duplet rule rather than the octet rule.	Ch5 §5.1; SLO (describe noble gas electronic configuration, octet and duplet rules ...)
39	The textbook gives three factors that determine the properties of a substance: the type of particles present, the way the particles are connected to	D. the arrangement of the particles 1	Section 5.5 lists the three factors as (1) type of particles (atoms, ions or molecules), (2) the way particles are	Ch5 §5.5; SLO (explain the properties of compounds in terms

	each other, and:		connected to each other, and (3) the arrangement of the particles.	of bonding and structure)
40	Metals can be hammered into thin sheets because, when a force is applied:	C. the layers of metal cations slide over one another 1	Section 5.6.1 explains that the layers in a metal are able to slide over each other without breaking the metallic bond, because the sea of delocalized electrons continues to hold the cations together. This makes metals malleable and ductile.	Ch5 §5.6.1; SLO (compare properties and use of materials such as strength and conductivity as determined by the type of chemical bond)
41	Ionic compounds are strong in compression but they are also:	C. brittle 1	Section 5.5.3 states that ionic compounds are strong in compression but brittle, meaning they break easily when a force is applied along the layers of ions.	Ch5 §5.5.3; SLO (compare properties and use of materials ... as determined by the type of chemical bond present)
42	Batteries and fuel cells make use of ionic compounds as:	D. electrolytes 1	Section 5.5.3 states that batteries and fuel cells use ionic compounds as electrolytes, because a molten or dissolved ionic compound contains freely moving ions that carry the current.	Ch5 §5.5.3; SLO (justify the availability of free charged particles for conduction of electricity in ionic compounds)
43	For which one of the following compounds are the empirical formula and the molecular formula exactly the same?	A. CH₄ 1	Section 6.1 lists methane among the compounds whose empirical and molecular formulas are identical. Glucose reduces to CH ₂ O, hydrogen peroxide to HO and ethane to CH ₃ .	Ch6 §6.1; SLO (define empirical formula as the simplest whole number ratio of different atoms in a molecule)
44	The formula mass of calcium hydroxide, Ca(OH) ₂ , is: (Ca = 40, O = 16, H = 1)	B. 74 amu 1	Formula mass = 40 + 2(16 + 1) = 40 + 34 = 74 amu.	Ch6 §6.2; SLO (use the relationship amount of substance = mass/molar mass to calculate ... relative formula mass)
45	The number of moles present in 1.5 kg of magnesium oxide, MgO, is: (Mg = 24, O = 16)	D. 37.5 mol 1	Molar mass of MgO = 24 + 16 = 40 g/mol. 1.5 kg = 1500 g. Moles = 1500 ÷ 40 = 37.5 mol.	Ch6 §6.5.1; SLO (use the relationship amount of substance = mass/molar mass to calculate number of moles)
46	The number of atoms present in 23 g of sodium is: (Na = 23)	B. 6.022 × 10²³ 1	23 g is the gram atomic mass of sodium, so it is exactly one mole. One mole contains Avogadro's number of atoms, 6.022 × 10 ²³ .	Ch6 §6.4; SLO (explain the relationship between a mole and Avogadro's constant)
47	When the equation ___ Ca + ___ H ₂ O → ___ Ca(OH) ₂ + ___ H ₂ is correctly balanced, the coefficients are:	C. 1, 2, 1, 1 1	Ca + 2H ₂ O → Ca(OH) ₂ + H ₂ . Left: 1 Ca, 4 H, 2 O. Right: 1 Ca, 2 + 2 = 4 H, 2 O. Balanced.	Ch6 §6.6.1; SLO (construct chemical equations to show reactants forming products)
48	Ions that appear unchanged on both sides of an ionic equation and take no part in the reaction are called:	D. spectator ions 1	Section 6.6.2 states that ions common to both sides do not actually take part in the chemical reaction and are called spectator ions. Removing them gives the net ionic equation.	Ch6 §6.6.2; SLO (construct chemical equations and ionic equations to show reactants forming products)
49	The molecular formula of the compound whose structural formula is CH ₃ -CH ₂ -NH ₂ is:	A. C₂H₇N 1	Count the atoms of each element: carbon 1 + 1 = 2, hydrogen 3 + 2 + 2 = 7, nitrogen 1. The molecular formula is therefore C ₂ H ₇ N.	Ch6 §6.7; SLO (deduce the formula of a molecular substance from a given structure of molecules)

50	The formula of the compound formed between calcium ions, Ca^{2+} , and phosphate ions, PO_4^{3-} , is:	C. $\text{Ca}_3(\text{PO}_4)_2$ 1	Balance the charges: $3(+2) = 2(-3)$, that is $+6 = -6$. So three calcium ions are needed for every two phosphate ions, and brackets are used because more than one polyatomic ion is present.	Ch6 §6.3; SLO (deduce the formula and name of a binary compound from ions given relevant information)
51	The binary ionic compound formed between aluminium and nitrogen is named:	D. aluminium nitride 1	Section 6.3 states that the cation is named first, keeping the name of the metal, and the mono-atomic anion takes the suffix -ide. Nitrogen therefore becomes nitride.	Ch6 §6.3; SLO (deduce the formula and name of a binary compound from ions given relevant information)
52	The mass of 1.5 moles of nitrogen gas, N_2 , is: (N = 14)	A. 42 g 1	Nitrogen is diatomic, so its molar mass is $2 \times 14 = 28$ g/mol. Mass = number of moles $\times$ molar mass = $1.5 \times 28 = 42$ g.	Ch6 §6.5.1; SLO (use the relationship amount of substance = mass/molar mass to calculate mass)
53	The number of molecules present in 0.5 moles of hydrogen peroxide, H_2O_2 , is:	A. 3.011×10^{23} 1	Number of molecules = number of moles $\times$ Avogadro's number = $0.5 \times 6.022 \times 10^{23} = 3.011 \times 10^{23}$ molecules.	Ch6 §6.5.2; SLO (use the relationship ... to calculate number of particles)
54	The oxidation number of nitrogen in nitric oxide, NO, is:	B. +2 1	Oxygen is -2 and the sum of the oxidation numbers in a neutral compound is zero. So $\text{N} + (-2) = 0$, giving $\text{N} = +2$.	Ch7 §7.2.2–7.2.3; SLO (explain that the sum of the oxidation numbers in a neutral compound is zero)
55	In the reaction $\text{CO}_2 + 2\text{Mg} \rightarrow 2\text{MgO} + \text{C}$, the reducing agent is:	A. Mg 1	Magnesium goes from 0 to $+2$, an increase, so it is oxidized. The reactant containing the element that is oxidized is the reducing agent, so Mg is the reducing agent and CO_2 is the oxidizing agent.	Ch7 §7.4.2; SLO (identify oxidizing and reducing agents in redox reactions)
56	In a polyatomic (complex) ion, the sum of the oxidation numbers of all the atoms present is equal to:	D. the charge on the ion 1	Rule 3 of Section 7.2.2 states that in a complex ion the sum of the oxidation states of the atoms is equal to the charge on the ion. For example, in CO_3^{2-} the sum is -2 .	Ch7 §7.2.2; SLO (explain that the sum of the oxidation numbers in an ion is equal to the charge on the ion)
57	In a redox reaction, the oxidizing agent or the reducing agent is:	A. the whole molecule or formula unit that contains the atom whose oxidation number changed 1	Section 7.4.2 states explicitly that the oxidizing or reducing agent is the whole molecule or formula unit and not the atom that has undergone the change in oxidation number.	Ch7 §7.4.2; SLO (identify oxidizing and reducing agents in redox reactions)
58	In the change $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$, the silver ion is:	B. reduced, because it gains an electron 1	Reduction is the gain of electrons. The silver ion accepts one electron and its oxidation state falls from $+1$ to zero, so it is reduced to metallic silver.	Ch7 §7.1.3; SLO (define redox reactions ... in terms of transfer of ... electrons and changes in oxidation state)
59	A collision between reacting particles can be effective only if the energy of the colliding particles is high enough to overcome the:	D. repulsion between the electrons around the reacting particles 1	Section 8.5 states that an effective collision can only occur if the energy of the colliding particles is high enough to overcome the repulsion between the electrons around the reacting particles, and if the orientation is correct.	Ch8 §8.5; SLO (define activation energy as minimum energy that colliding molecules must have for a successful collision)
60	A catalyst speeds up a reaction by providing a new mechanism of lower activation energy, but it does not change the:	A. overall enthalpy of the reaction 1	Section 8.6 states that a catalyst increases the rate of a reaction by lowering its activation energy, but it does not affect the overall thermodynamics or enthalpy of	Ch8 §8.6; SLO (explain that activation energy depends on reaction pathway which can be changed using

SECTION B — Short Questions (2 × 3 = 6 Marks)

Q.2 (i) A compound of nitrogen and oxygen contains 30.4% nitrogen and 69.6% oxygen by mass; relative molecular mass = 92. (a) Find the empirical formula. (b) Find the molecular formula. 3

(a) Empirical formula 2

Take 100 g of the compound, so the percentages become masses in grams.

Moles of N = $30.4 \div 14 = 2.17$ Moles of O = $69.6 \div 16 = 4.35$ **1**

Divide both by the smaller value, 2.17:

N = $2.17 \div 2.17 = 1$ O = $4.35 \div 2.17 = 2$ (2.004, rounded to 2)

Simplest whole-number ratio N : O = 1 : 2, so the **empirical formula is NO₂** **1**

(b) Molecular formula 1

Empirical formula mass of NO₂ = $14 + 2(16) = 46$

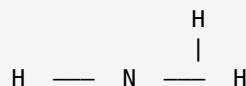
$n = \text{relative molecular mass} \div \text{empirical formula mass} = 92 \div 46 = 2$

Molecular formula = (NO₂)₂ = **N₂O₄** **1**

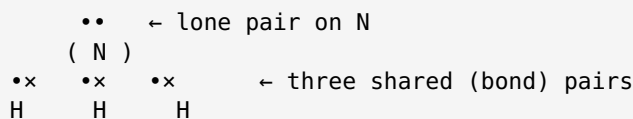
Marking note: allow full credit if she divides by 2.17 and writes 1 : 2 directly. Award the mark in (b) even if (a) is wrong, provided her own empirical formula is used correctly.

Q.2 (ii) Ammonia, NH₃. (a) Draw a dot-and-cross diagram to show the bonding in one molecule. (b) State the number of bond pairs and lone pairs around the nitrogen atom. 3

(a) Dot-and-cross diagram 2



Dot-and-cross (N electrons = • , H electrons = ×)



N (Z = 7) = 2, 5 → 5 valence electrons, needs 3 more to complete its octet.

Each H (Z = 1) has 1 electron and needs 1 more to complete its duplet.

N therefore shares one electron with each of the three H atoms:

3 bond pairs are formed and 1 pair on nitrogen is left unshared.

1 mark for showing nitrogen with three shared pairs, each shared pair containing one dot and one cross.

1 mark for showing the unshared (lone) pair on nitrogen and for giving each hydrogen two electrons.

(b) Around the nitrogen atom there are 3 bond pairs (one to each hydrogen atom) and 1 lone pair. 1

Marking note: accept “3 shared pairs and 1 unshared pair”. Do not deduct if she draws the three N-H bonds as plain lines in part (a), provided the lone pair is shown.

SECTION C — Long Question (1 × 5 = 5 Marks)

Q.3 Potassium sulphate, K_2SO_4 . (K = 39, S = 32, O = 16, H = 1) 5

(a) Formula mass of K_2SO_4 1

K: $2 \times 39 = 78$ S: $1 \times 32 = 32$ O: $4 \times 16 = 64$

Formula mass = $78 + 32 + 64 = 174 \text{ amu}$ (molar mass = 174 g/mol)

(b) Number of moles in 60.9 g 1

Number of moles = mass $\div$ molar mass = $60.9 \div 174 = 0.35 \text{ mol}$

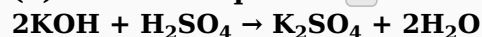
(c) Number of formula units 1

Number of formula units = number of moles $\times$ Avogadro's number

= $0.35 \times 6.022 \times 10^{23} = 2.108 \times 10^{23} \text{ formula units}$

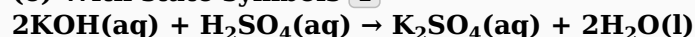
(Accept 2.11×10^{23} . K_2SO_4 is ionic, so the particles are *formula units*, not molecules.)

(d) Balanced equation 1



Coefficients: **2, 1, 1, 2**. Check — left: 2 K, 6 O (2 + 4), 4 H, 1 S; right: 2 K, 6 O (4 + 2), 4 H, 1 S. Balanced.

(e) With state symbols 1



(aq) = dissolved in water, (l) = pure liquid. Award the mark only if all four state symbols are correct and the equation is still balanced.

Marking note: carry her own value forward. If (a) is wrong but (b) and (c) use her value correctly, award (b) and (c) in full.

Answer-letter balance: 15 A, 15 B, 15 C, 15 D across the 60 MCQs.

Chapters covered: Chemistry Class 9 — Chapters 1 to 9.

— End of Answer Key —